Digital Biquad Filter with Software Implementation For ATM3330e/ATM3330 Application Note

Revision History

Date	Version	Description
January 31, 2022	0.50	Initial version created.

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1 Overview

This document introduces the software implementation of a digital biquad filter integrated with the HID_remote application. It will also present the actual frequency response and some software metrics of an example biquad filter measured with the ATM3330e/ATM3330 (ATM33) platform under various gain settings.

1.1 References

Description	Location	Notes
CMSIS DSP Software Library	https://arm-software.github.io/CMSIS_5/DSP/html/index.html	See Tab - Reference/ Examples/ Graphic Audio Equalizer Example

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2 Introduction

A digital biquad filter is a second order IIR filter that is widely used in signal processing to vary the tonal quality of the audio. Considering that the higher order filters also have higher sensitivity of their coefficients, the biquads are often cascaded to build more complex filters for stability.

The CMSIS-DSP software library provides a suite of common signal processing functions, including the filtering functions, for use on Cortex-M and Cortex-A processor based devices. Although the library is released in source form, it is already prebuilt for all cores and integrated with some IDE such as IAR EW or Keil uvision. In that case, the library could be used directly without extra effort needed.

Based on different coefficients configured, the biquads could be used to implement any of the filter types such as low-pass, high-pass, band-pass, all-pass, notch, peak, shelf, etc., depending on the application needs. For illustration, a graphic audio equalizer example provided in the CMSIS-DSP library is used to demonstrate how a 5-band equalizer can be constructed by cascaded biquad functions. This example is referenced and integrated with the HID_remote application to measure different metrics influenced by the filter.

3 Information About the Equalizer Example

The design is based on a cascade of 5 filter sections. Each section is 4th order and consists of two biquads in direct form 1 which better fits the fixed point audio data for this example. See Figure 1 (sourced from https://arm-software.github.io/CMSIS_5/DSP/html/GEQ_signalflow.gif).

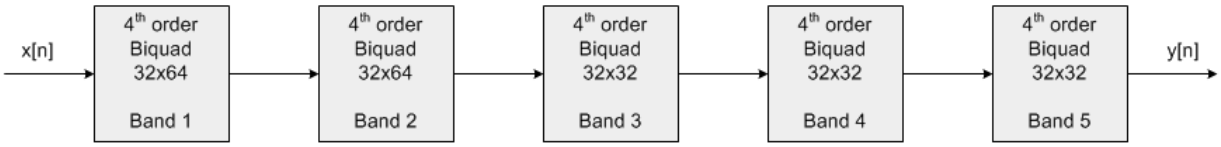


Figure 1 - Cascaded Filters

Each filter section has a nominal gain of 0 dB and can be separately set to boost or cut the input signal within its specific frequency range. The example already has a precomputed coefficient table for each filter to boost or cut signals in the range +/- 9 dB with 1 dB steps.

The frequency response of these 19 different settings for the 5 filters is shown in Figure 2 (sourced from https://arm-software.github.io/CMSIS_5/DSP/html/GEQ_allbandresponse.gif).

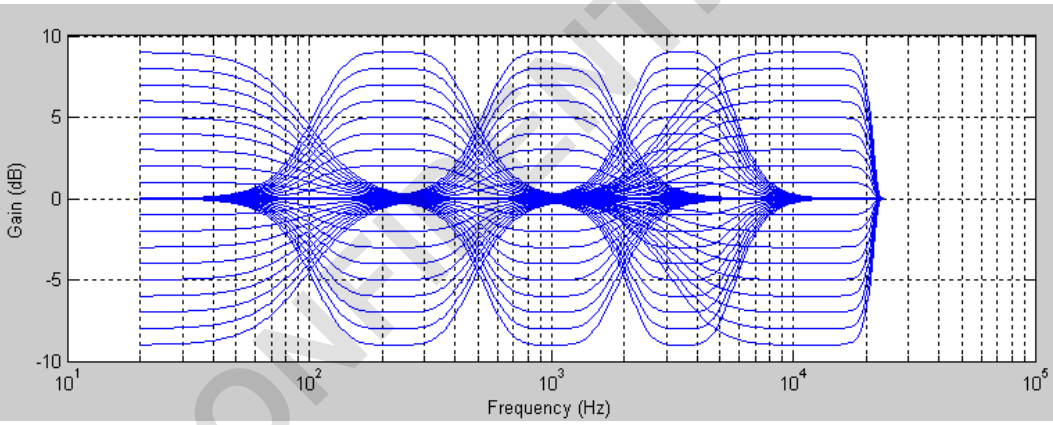


Figure 2 - Filters' Frequency Response

In order to get better noise performance, the original implementation uses the high precision 32x64-bit biquad filters for the first two bands and standard 32x32-bit biquad filters for the remaining 3 bands. Since the input/output data of the biquads is in Q31 format, a converting stage for the ATM33 Q15 format PCM data is therefore necessary.

The effect of the equalizer is usually verified by using a logarithmic chirp as its input. Since such a signal increases its frequency with time, one can easily examine if the signal in a specific frequency is boosted or cut as expected. The CMSIS-DSP library gives an example with the gain level for the five bands set to (0, -3, 6, 4, -6) respectively.

The original input signal before the equalizer is shown in Figure 3 (sourced from https://arm-software.github.io/CMSIS_5/DSP/html/GEQ_inputchirp.gif).

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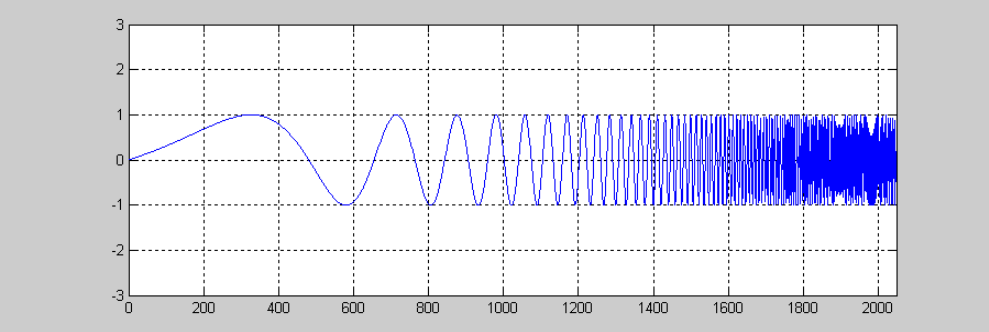


Figure 3 - Output Signal Before the Equalizer

After processed by the equalizer with gain level set to (0, -3, 6, 4, -6), the output signal will be as shown in Figure 4 (sourced from https://arm-software.github.io/CMSIS_5/DSP/html/GEQ_outputchirp.gif).

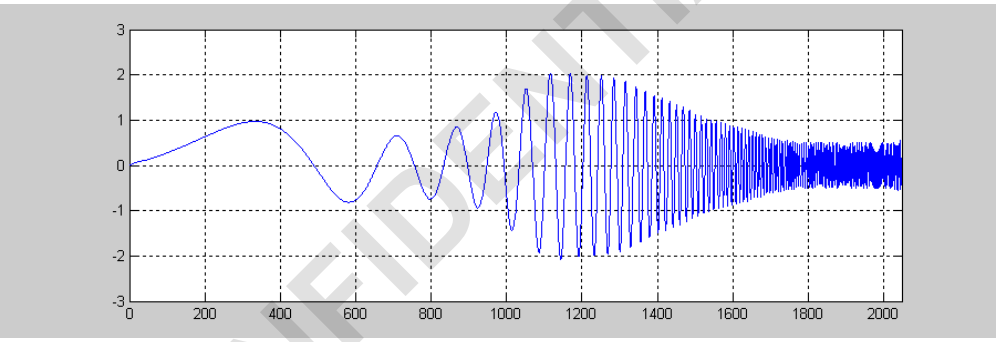
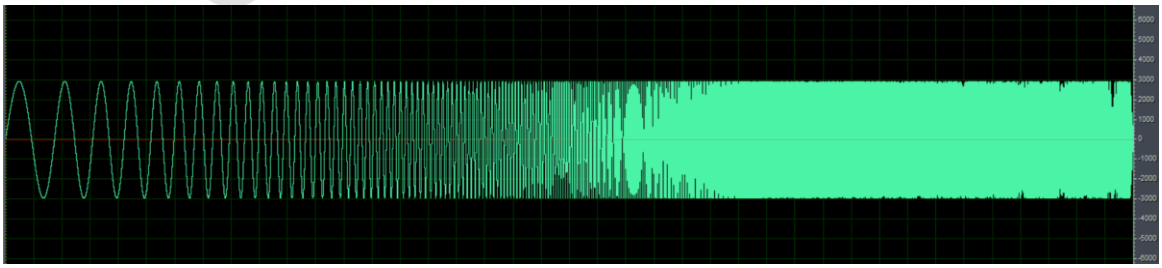


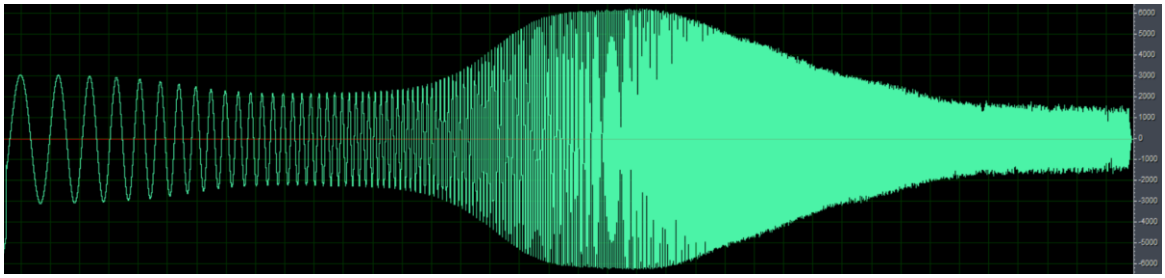
Figure 4 - Output Signal After the Equalizer

Similar test is performed with ATM33 by using the following chirp as the input:
(sweep from 10 Hz to 6000 Hz in 2 seconds with -21 dBFS strength)



This chirp signal is pre-made and embedded in software as a 16-bit PCM form with sampling rate of 16 KHz. Once the build option introduced in the Makefile Settings section is enabled, the biquad module would replace the actual input data from the PDM module with the fixed chirp tone periodically. This way any acoustic gain of the external device will not be involved. After processed by the equalizer with the gain level also set to (0, -3, 6, 4, -6), the output is as expected:

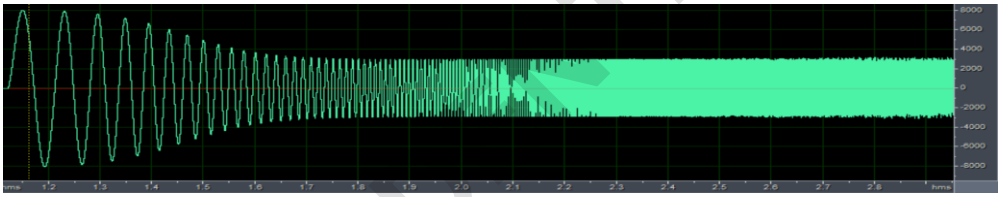
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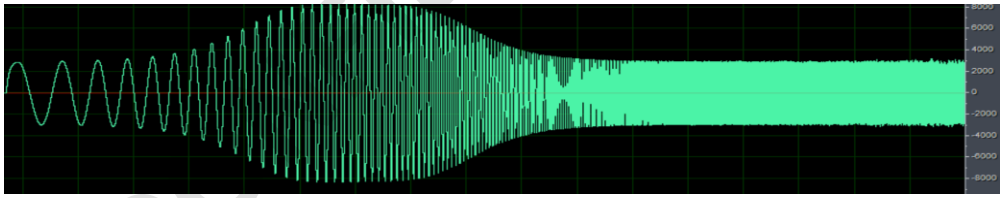
4 Frequency Response

Other combinations of the gain level are used, here are the results:

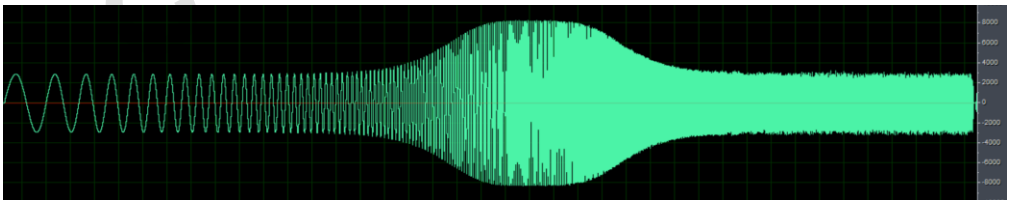
gain set to (9, 0, 0, 0, 0)



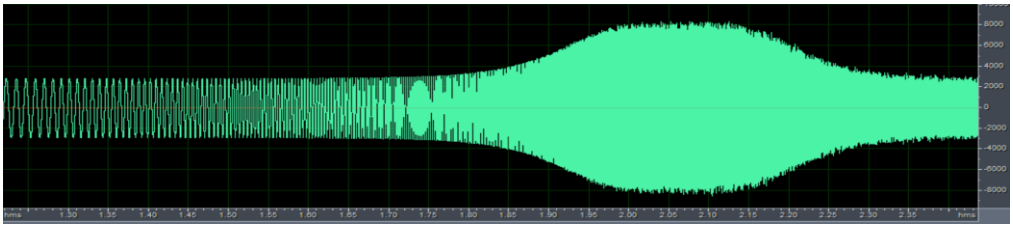
gain set to (0, 9, 0, 0, 0)



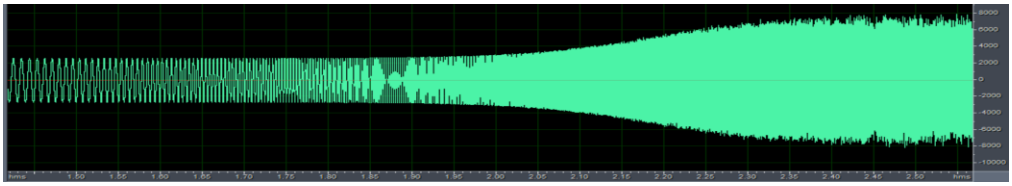
gain set to (0, 0, 9, 0, 0)



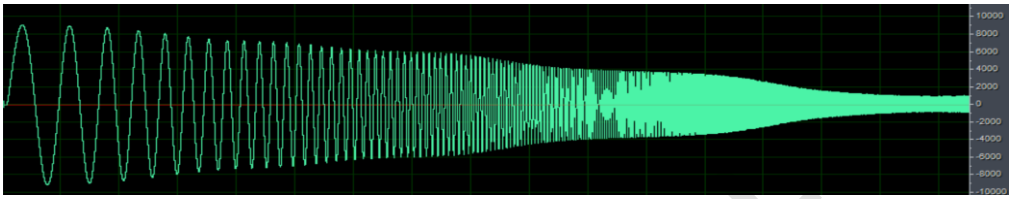
gain set to (0, 0, 0, 9, 0)



gain set to (0, 0, 0, 0, 9)



gain set to (9, 5, 1, -4, -9)



5 System Profiling

5.1 CPU Utilization

For this experiment, the program on the Cortex-M33 processor is running only at 16 MHz. As the original design uses two high precision 32x64-bit biquad filters in lower frequency bands for better noise performance, the CPU utilization is certainly higher than just using standard 32x32-bit. Therefore the tradeoff between performance, computing resources and power consumption must be considered.

The frequency response presented in the previous section is measured with the same design as the original GEQ example which uses two 32x64-bit biquads in the application. Different configurations such as changing each band to 2nd order, changing the number of bands in the process or using only 32-bit biquads and measuring this metric are experimented. The CPU utilization is calculated based on the elapsed time of the entire biquad process over the PCM interrupt period (2 ms). Since the biquad process might be interrupted by other tasks during a Bluetooth LE connection, the number might vary within a certain range. To better understand the distribution, the max/min/avg value of the CPU utilization for each case is recorded. The execution time of the audio encoder is not included in this measurement.

Description of all band	Order of Each Band	Maximum CPU Utilization	Average CPU Utilization	Minimum CPU Utilization
32x32 * 1	2	15%	7%	4%
32x64 * 1	2	25%	13%	9%
32x32 * 3	2	21%	12%	9%

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32x32 * 5	2	25%	17%	13%
32x64 * 2 + 32x32 * 3	2	45%	27%	22%
32x32 * 5	4	42%	25%	22%
32x64 * 2 + 32x32 * 3	4	68%	48%	41%

Table 1 - CPU Utilization of the Biquad Process at 16 MHz

5.2 RAM Usage

Since the biquads are cascaded, the filter functions are also executed sequentially in the processing stage hence the heap usage does not increase with the order of the filter. The static RAM usage for each extra filter would just be a state buffer that holds the value of the previous 4 data points and a static instance of the filter configuration.

- 32x64-bit filter: 32-bytes buffer + 16-bytes instance structure.
- 32x32-bit filter: 16-bytes buffer + 16-bytes instance structure.

6 Coefficients for the Biquad Filter

This section describes how to configure the coefficients for each biquad filter stage.

Each biquad stage is described by 5 feedforward/feedback precomputed constants that are usually denoted by b0, b1, b2, a1, a2 and the actual implementation of CMSIS DSP library is using the difference equation:

```
y[n] = b0 * x[n] + b1 * x[n-1] + b2 * x[n-2] + a1 * y[n-1] + a2 * y[n-2]
```

Note that some design tools might be using another difference equation with minus signs:

```
y[n] = b0 * x[n] + b1 * x[n-1] + b2 * x[n-2] - a1 * y[n-1] - a2 * y[n-2]
```

In that case the feedback coefficient a1 and a2 obtained by such a tool must be negated when used with the CMSIS DSP Library.

Once the coefficients for a biquad stage are decided, store them in an array with the same datatype as the input. For example, use q31 coefficients for q31 biquads and use f16 coefficients if the f16 version is selected. The coefficients are stored in the following order:

{b0, b1, b2, a1, a2}

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If there are going to be multiple stages for that biquad filter with a single function call, append the coefficients sequentially and specify the number of stages in the API parameter:

{b10, b11, b12, a11, a12, b20, b21, b22, a21, a22, ...}

where b1x/a1x stands for the coefficients for the first stage, b2x/a2x are for the second stage, and so on.The coefficient array is then provided as an input parameter of the initialization function for each data type.

For more detailed information, please visit the following URL: https://arm-software.github.io/CMSIS_5/DSP/html/group__BiquadCascadeDF1.html

NOTE: Although the fixed-point Direct-Form 1 function is used for illustration, the above can be extended to Direct-Form 2-transposed structures as well. Please refer to the following URL for additional details:
https://arm-software.github.io/CMSIS_5/DSP/html/group__BiquadCascadeDF2T.html

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7 Makefile Settings

If the build process is triggered by makefile rather than an IDE, some build variable must be passed to the make process.

Use CFG_BIQUAD_ENABLE=1 to enable the biquad process.

Use CFG_BIQUAD_SWEEP_TEST=1 to replace the actual PCM data with the internal chirp tone.

Use CFG_CMSIS_DSP_LIB_NOT_FROM_IDE=1 to use an external CMSIS_DSP library. Should also modify the include path if that's different from the default.

Use CFG_BIQUAD_IN(OUT)_SCALE=[desired value] to set the shift factor before/after the biquad process to provide additional headroom so that the graphic EQ may apply gain.



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